Predicting Graft Acceptance/Rejection After Liver Transplantation

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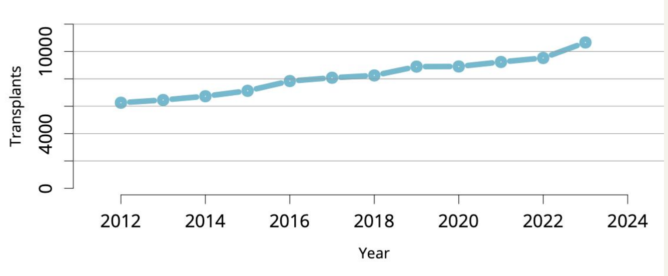
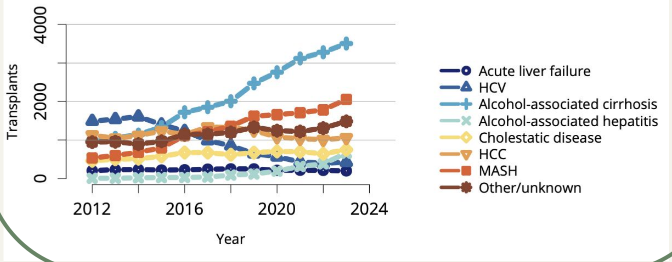
Background: Liver Transplantation and the Need for Predictive Diagnostics

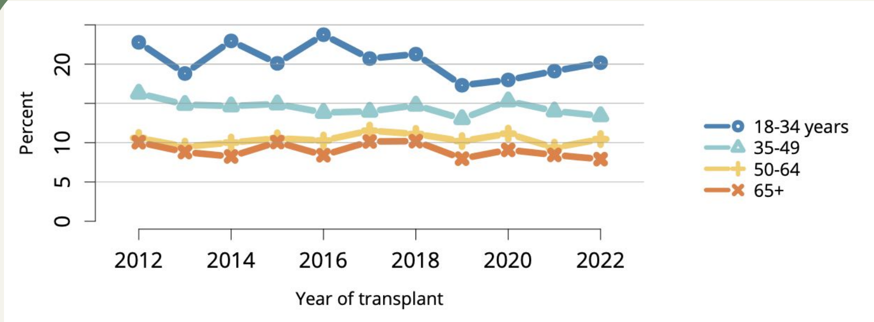
"In 2023, liver transplants in the United States reached a new record, totaling 10,659 procedures — 10,125 (95%) in adults and 534 (5%) in children." [1]

Overall liver transplants



Adult liver transplants by diagnosis.





Incidence of acute rejection by 1 year post-transplant among adult liver transplant recipients by age!!

[1] Kwong, Allison J., et al. "OPTN/SRTR 2023 annual data report: liver." American Journal of Transplantation 25.2 (2025): S193-S287.

Background: Liver Transplantation and the Need for Predictive Diagnostics

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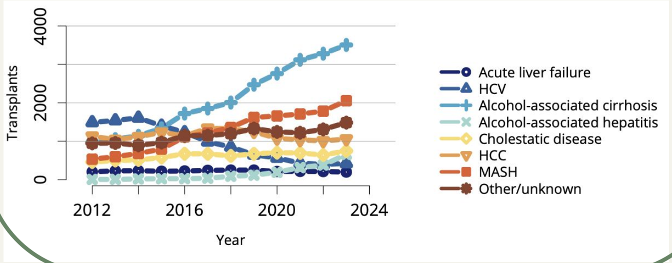
Overall liver transplants

Current method of detecting rejection: *Traditional Histopathology (Banff Criteria)*

Limited by inter-observer variability, Pathologist-dependent, Subjective, labor-intensive

Adult liver transplants by diagnosis.



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Goal:

*Build a computational, **gene expression–based predictive model** that distinguishes **rejection** from **non-rejection liver transplant biopsies**, enhancing diagnostic accuracy beyond traditional histopathology.*

Data source, classification of molecular genotypes

- As part of their ongoing clinical study, Madill-Thomsen, Katelynn S., et al. [2] performed transcriptome profiling on 763 liver transplant biopsies, identifying four distinct molecular genotypes.
    - *No-rejection 54%,*
    - *TCMR (T-cell mediated rejection) 16%,*
    - *NKRL (NK cell–enriched rejection-like) 13%,*
    - *Injury 16%.*

- The gene expression dataset is publicly available on the Gene Expression Omnibus (GEO) website under accession number GSE277334.

- For our predictive model, we classified the samples into 'rejection' and 'non-rejection' groups by clustering based on their molecular genotypes as follows:

Non-Rejection (0)	**Rejection (1)**
No rejection	TCMR
Injury	NKRL

[2] Madill-Thomsen, Katelynn S., et al. "Defining an NK Cell–enriched Rejection-like Phenotype in Liver Transplant Biopsies From the INTERLIVER Study." Transplantation (2025): 10-1097.

Data Analysis Workflow

Project roadmap

Step 1

Data Exploration

-Data sources

Step 2

Preprocessing

-Data Cleaning

-Data labelling

-Normalization

Step 3

Exploratory Data Analysis (EDA)

-Feature engineering

-Feature selection

Step 4

Modelling

-Model training

-Performance metrics

Step 5

Summary of Findings

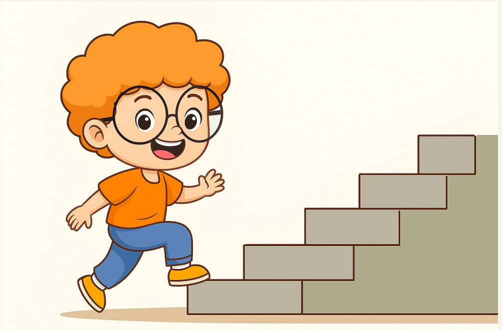
-Accuracy score

-ROC-AUC score

-Standard deviation

Next steps

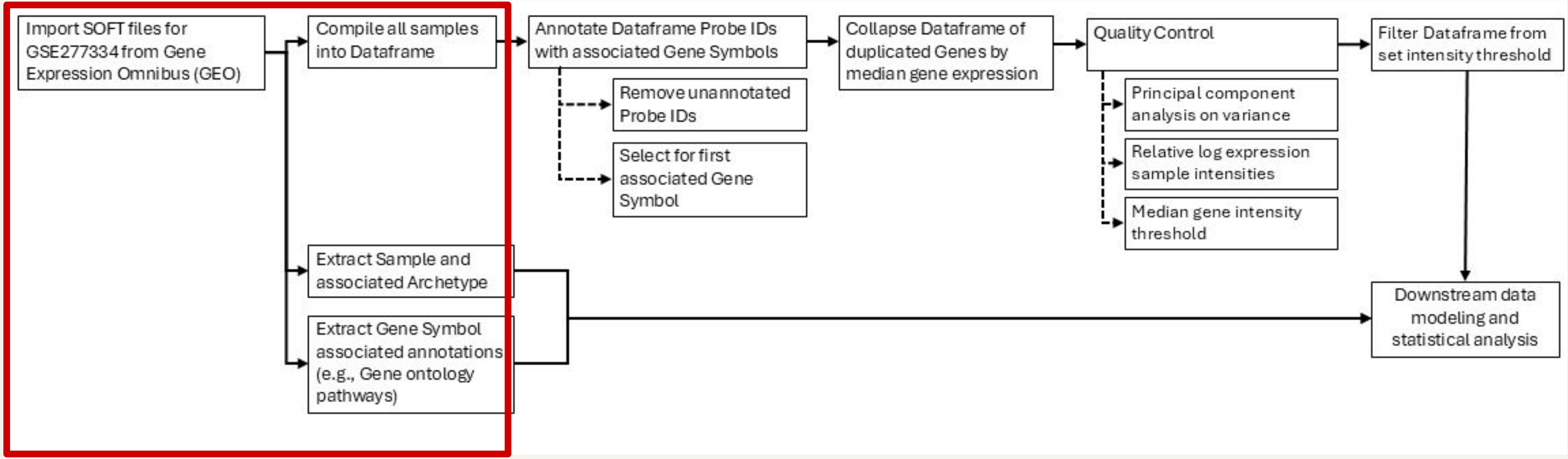
Diagnostic predictive tool



Microarray data processing and cleaning

Patient samples: 763, Non-rejection (415), T cell-mediated rejection (TCMR, 125), Injury (125), and NK cell-enriched rejection-like state (NKRL, 98)

Gene features filtering: 48774 (initial) ->



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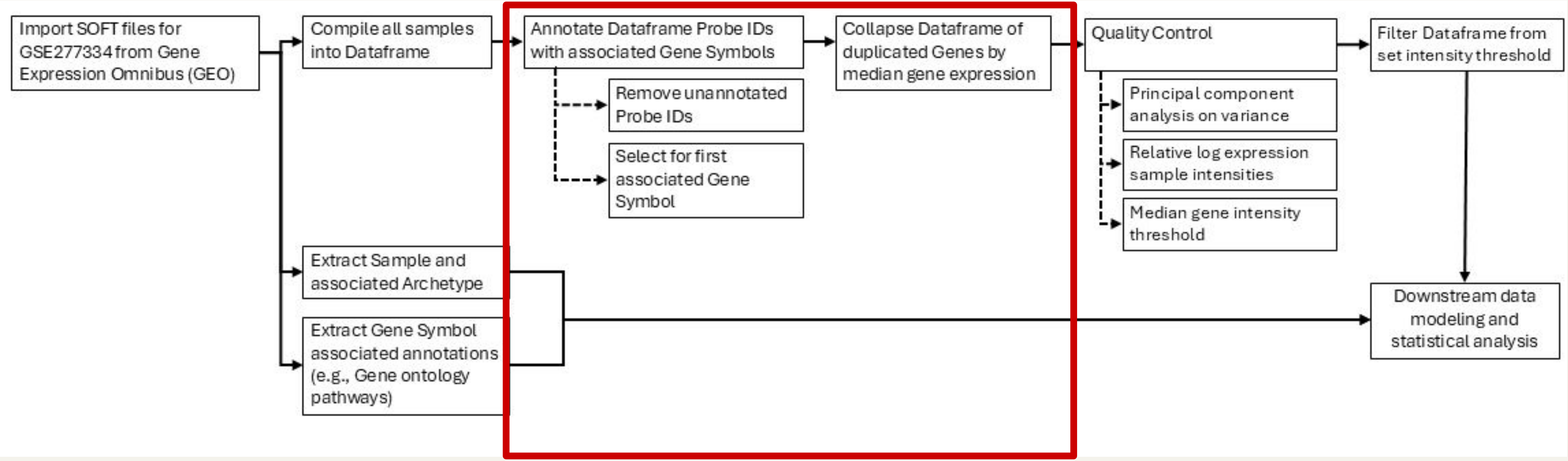

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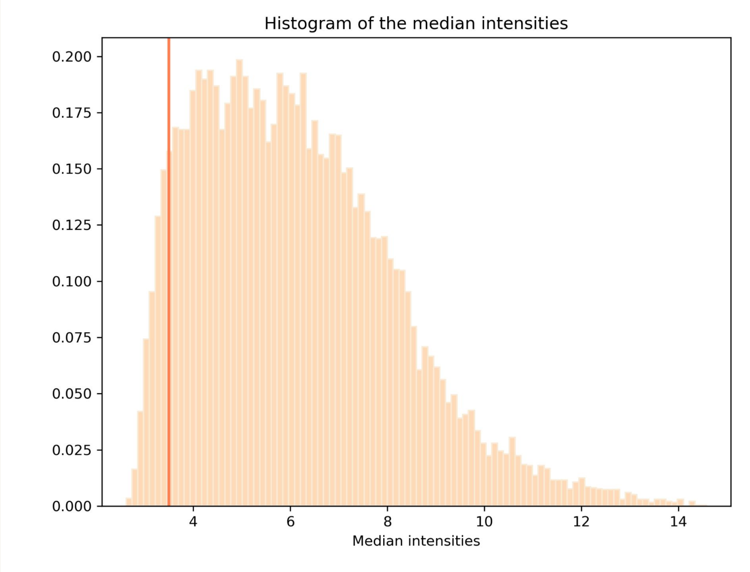

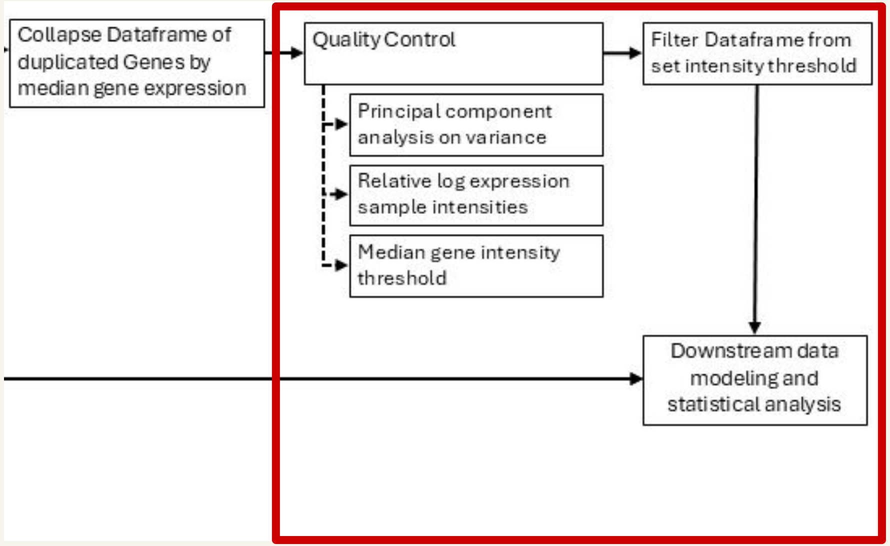

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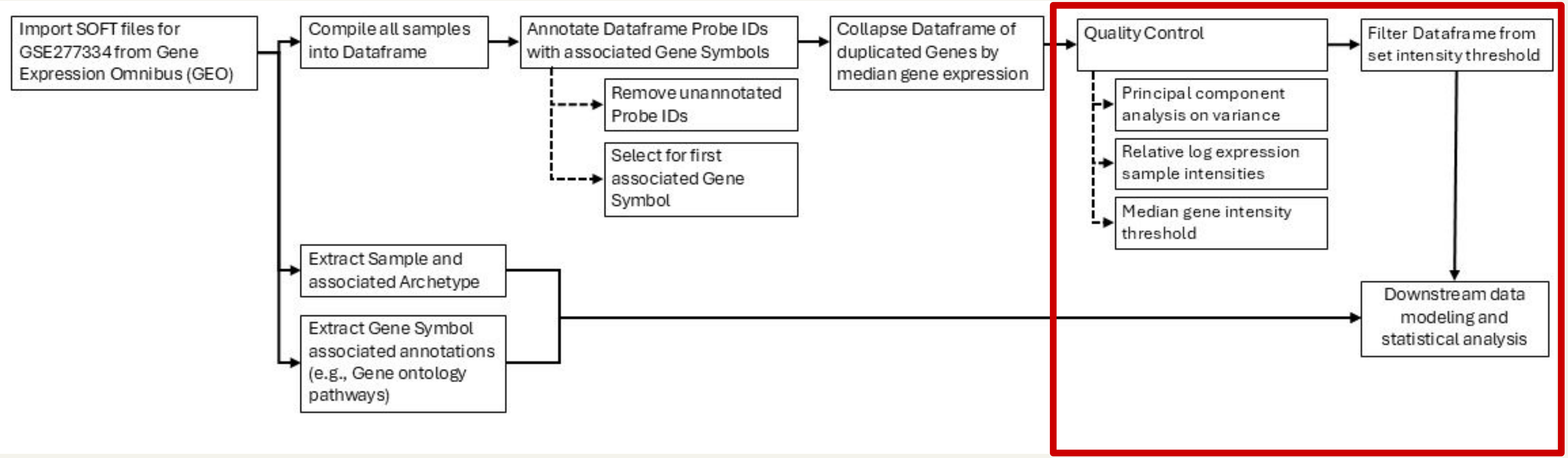
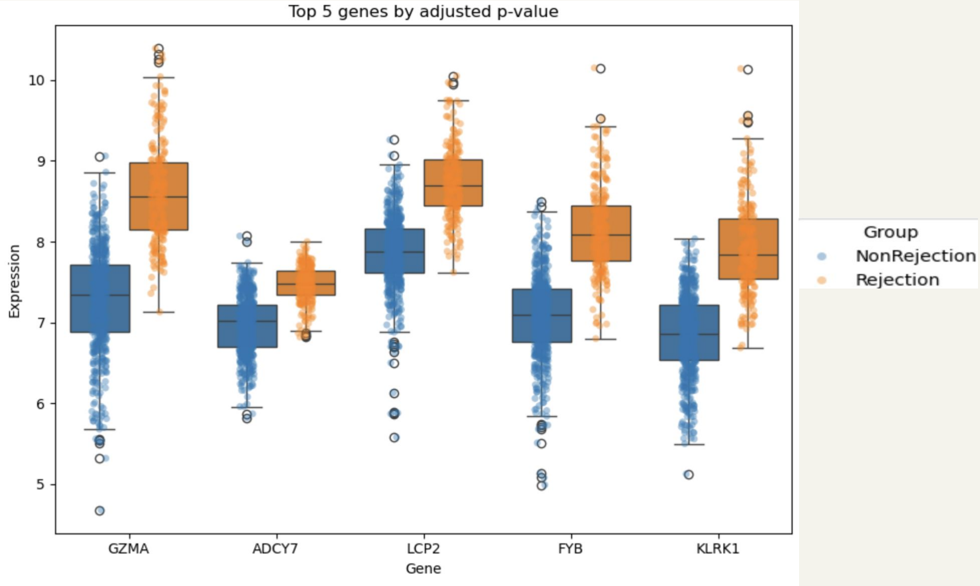
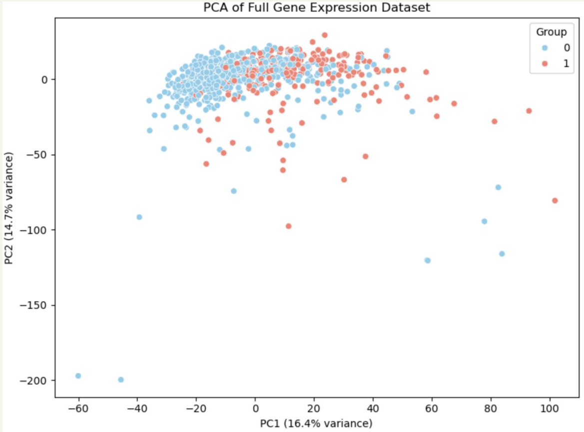


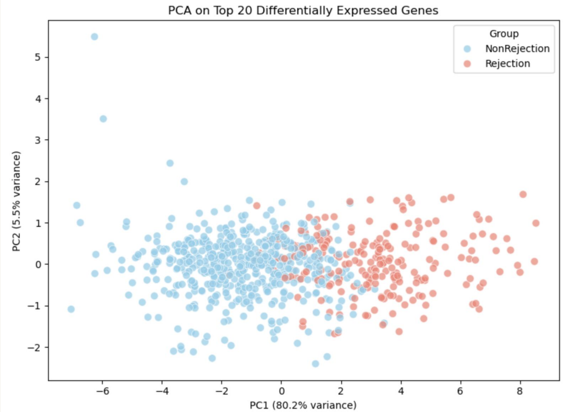
Exploratory Data Analysis: ***PCA and t-test***



PCA on all the genes

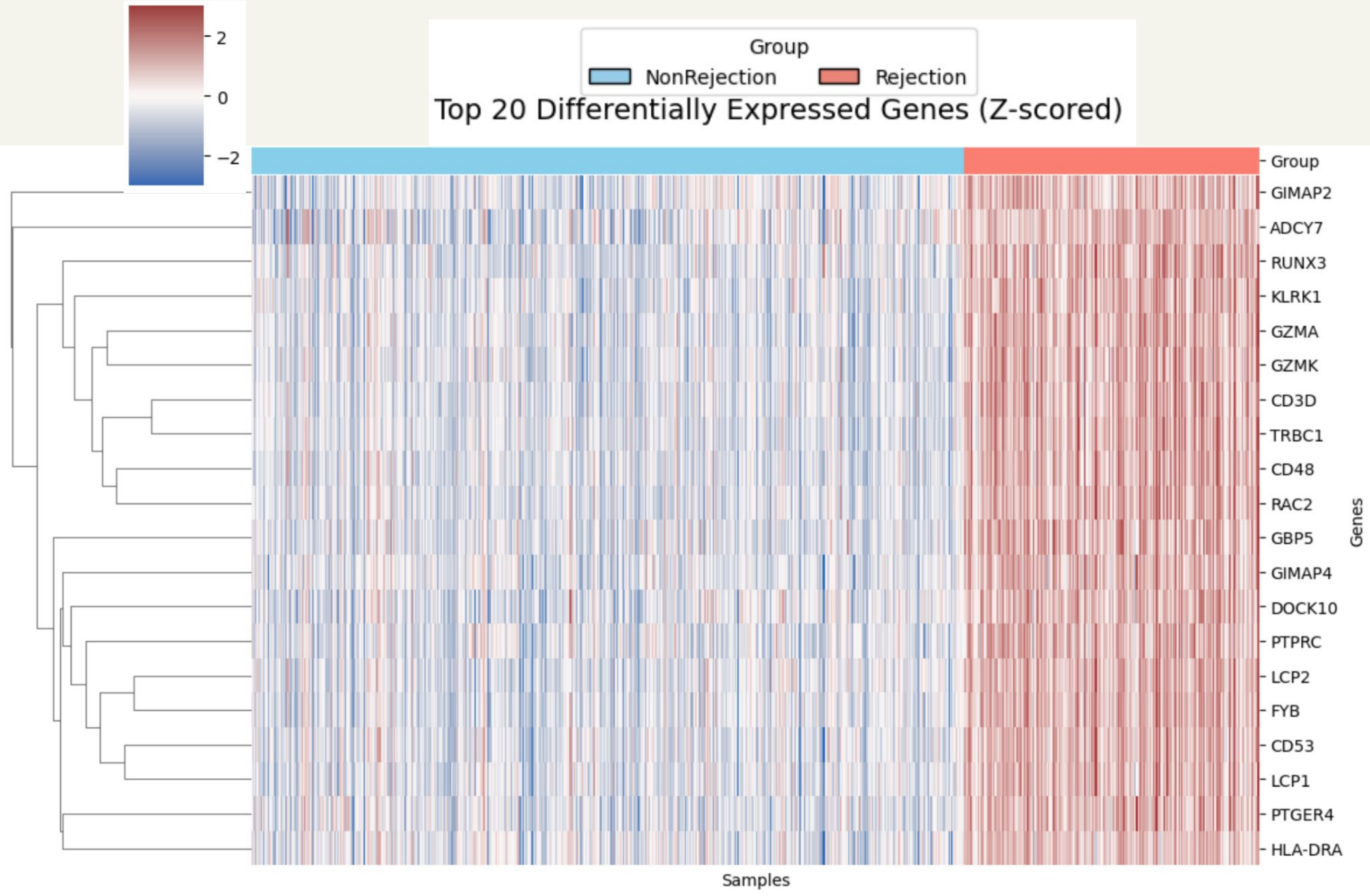


PCA on top 20 genes selected from t-test



Gene	*Function*
GZMA	GZMA encodes Granzyme A, a cytotoxic protease released by T cells and NK cells to induce target cell death.
ADCY7	Influences cytokine production, cell activation, and immune signaling thresholds.
LCP2	T cell development and activation: critical for proper adaptive immune responses.
FYB	Critical for T cell activation and immune synapse formation.Influences cytotoxic T cell and NK cell responses.
KLRK1	NK cell–mediated killing of virus-infected or tumor cells.Co-stimulation of CD8+ T cells for enhanced cytotoxic activity.

Exploratory Data Analysis: Heat-map and interpretation of overexpressed genes



<u>Key functional categories:</u>

Cytotoxic and immune effector activity

GZMA, GZMK, KLRK1 – markers of cytotoxic T cells and NK cells.

FYB, LCP2, LCP1, RAC2 – T cell signaling and activation.

T cell and lymphocyte markers

CD3D, TRBC1, PTPRC, CD48, CD53 – T cell and lymphocyte identity and signaling.

Antigen presentation & immune regulation

HLA-DRA, PTGER4, ADCY7, GBP5 – immune signaling, inflammation, and antigen processing.

Other immune modulators / regulators

DOCK10, GIMAP2, GIMAP4, RUNX3 – regulate lymphocyte survival, differentiation, and trafficking.

Feature Selection

t-test

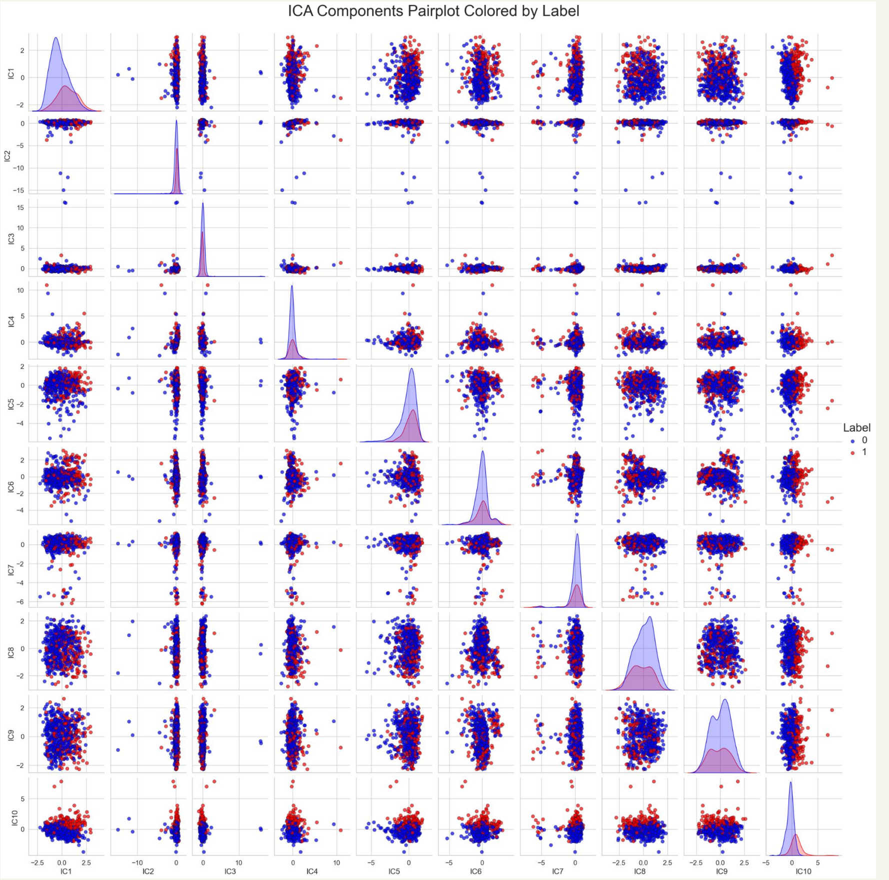
- Identifies the most statistically relevant genes that contributes to distinguishing between classes

Principal Component Analysis

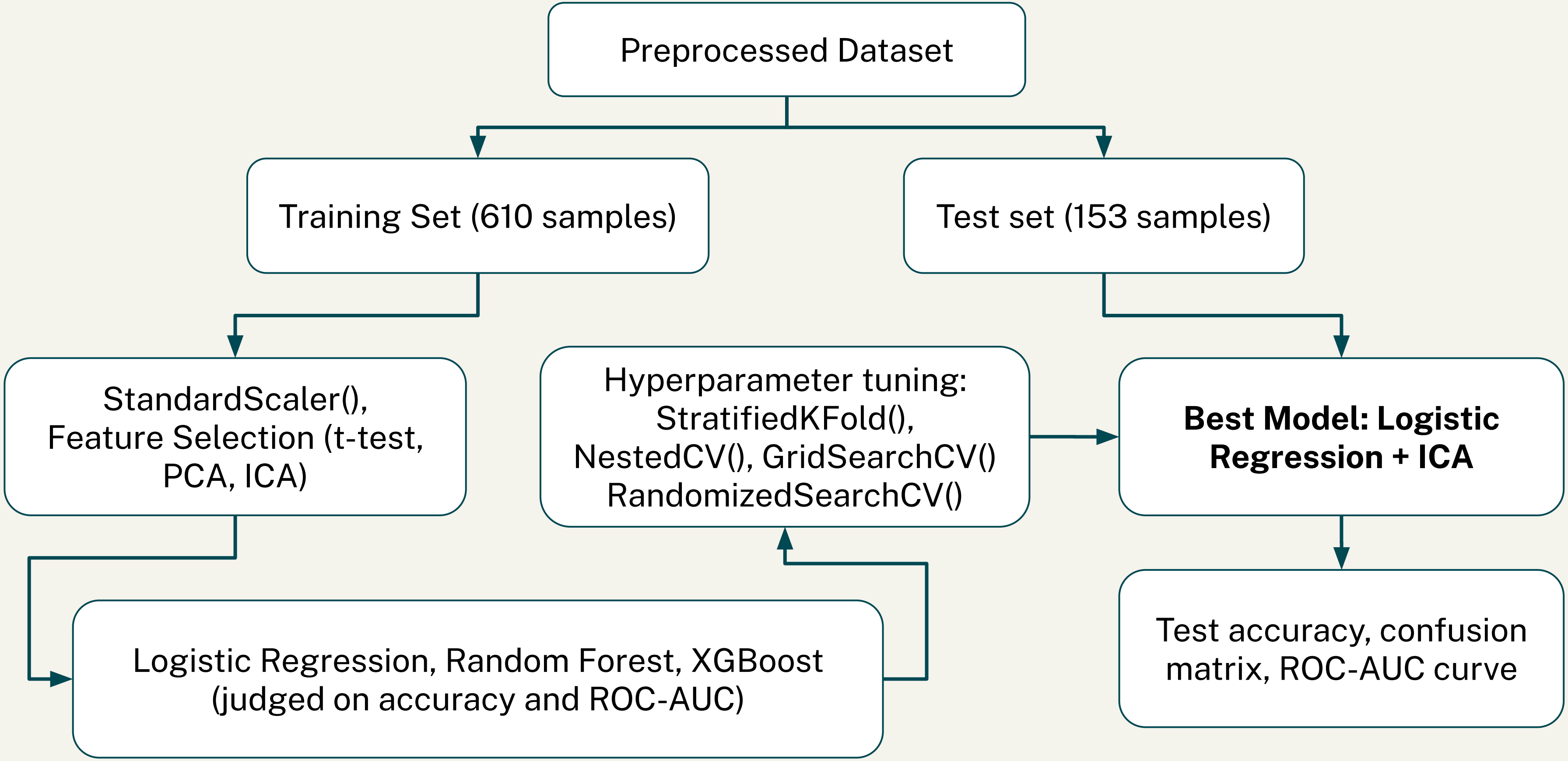
- Identifies components with maximum variance

Independent Component Analysis (ICA)

- Method in signal processing used to separate mixed signals into additive subcomponents
- Decomposes the dataset into components that are as statistically independent from the other as possible
- Useful in capturing latent signals that potentially reveal underlying gene expression patterns for gene clustering
- Can apply a model on the components directly, or **extract the features that contribute the most to a distinct component**



Modeling Workflow



Results

Modeling and Validation

- All of our models have high training accuracy and ROC-AUC scores
- **Best model: Logistic Regression + ICA Reduced (500 genes)**

Model	Accuracy	ROC-AUC
Logistic Reg + Full Dataset	0.92295	--
Logistic Reg + Reduced (1000)	0.90820 ± 0.01410	0.96923 ± 0.01004
Logistic Reg + Reduced (400)	0.9 ± 0.02034	0.96576 ± 0.00890
Logistic Reg + PCA	0.89016 ± 0.02464	0.94577 ± 0.00868
Logistic Reg + ICA	0.89180 ± 0.02224	0.94732 ± 0.00940
Logistic Reg + ICA Reduced (500)	**0.91803 ± 0.01719**	**0.97306 ± 0.01061**
Random Forest + Reduced (1000)	0.90984 ± 0.01639	0.96548 ± 0.00907
XGBoost + Reduced (1000)	0.91475 ± 0.01332	0.97041 ± 0.00700

Modeling and Validation

Logistic Regression + ICA Reduced (500 genes)

- **Overall accuracy: 0.88889**, **AUC score: 0.961**
  - Does well in classifying the non-rejection group with **high specificity**
  - The rejection group has **good recall** but still has a **false negative rate of ~24%**

```
Test Accuracy: 0.8888888888888888

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.94      0.92       108
           1       0.85      0.76      0.80        45

    accuracy                           0.89       153
   macro avg       0.88      0.85      0.86       153
weighted avg       0.89      0.89      0.89       153
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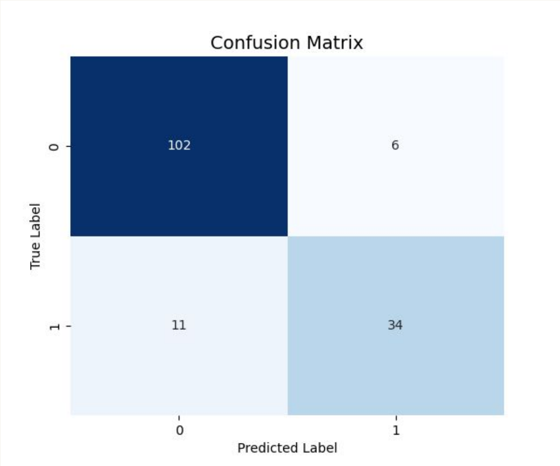
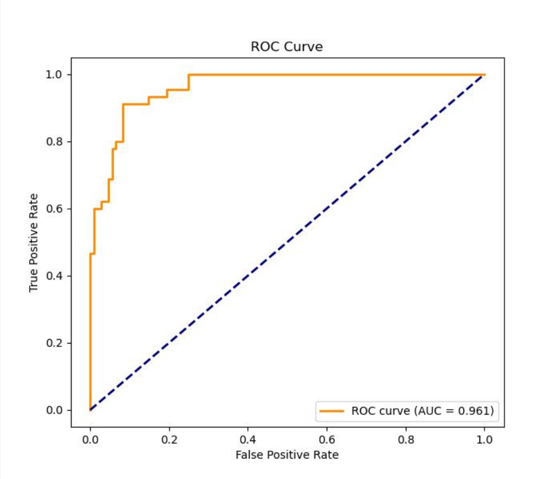
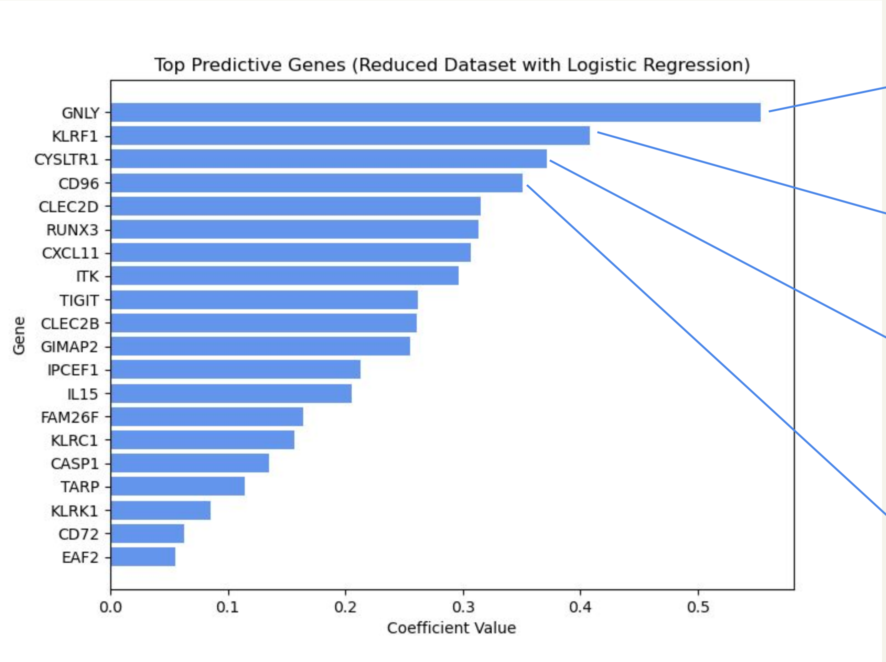




Modeling and Validation

Logistic Regression + ICA Reduced (500 genes)

- Top predictive genes from our model
  - Corresponds to **activated natural killer (NK) cells** and **cytotoxic CD8$^+$ T cells** for immune response, leading to rejection



Produces a protein is present in cytotoxic granules of cytotoxic T lymphocytes and natural killer (NK) cells

Expressed on nearly all NK cells and stimulates their cytotoxicity and cytokine release

Expressed on the outer membrane of immune/inflammatory cells and structural cells

May involve adhesive interactions of activated T and NK cells during the late phase of the immune response

Conclusions

Clinical impact, performance and future directions

- Clinical Impact
  - Our model captures a biologically relevant signal — the immune activation underlying rejection to assist rejection diagnostic accuracy

- Model Performance
  - Logistic Regression + ICA Reduced (500 genes) model had the highest accuracy and ROC-AUC scores
    - Modestly high false negative rate
    - Large number of features dropped

- Future Directions
  - Increase in biopsie patient cohort
  - Additional feature selection tuning
  - Inclusion of clinical patient measurements (e.g., histology, liver function tests)

Thank You!